

StayCertain



EVERY YEAR, HOTELS LOSE THOUSANDS OF ROOM-NIGHTS
TO CANCELLATIONS NO ONE SAW COMING.

Turning booking data into earlier warnings — and smarter decisions.

A decision-support tool for hotel revenue and front-desk teams

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THE PROBLEM

Cancellations are invisible — until it's too late.

75.8%

of hotel bookings
in our study were cancelled

*Nearly 1 in 3 reservations — with no
early warning at all.*



Lost Revenue

Rooms sit empty with no time to resell them.



Broken Forecasts

Occupancy and staffing plans are built on numbers that quietly erode.



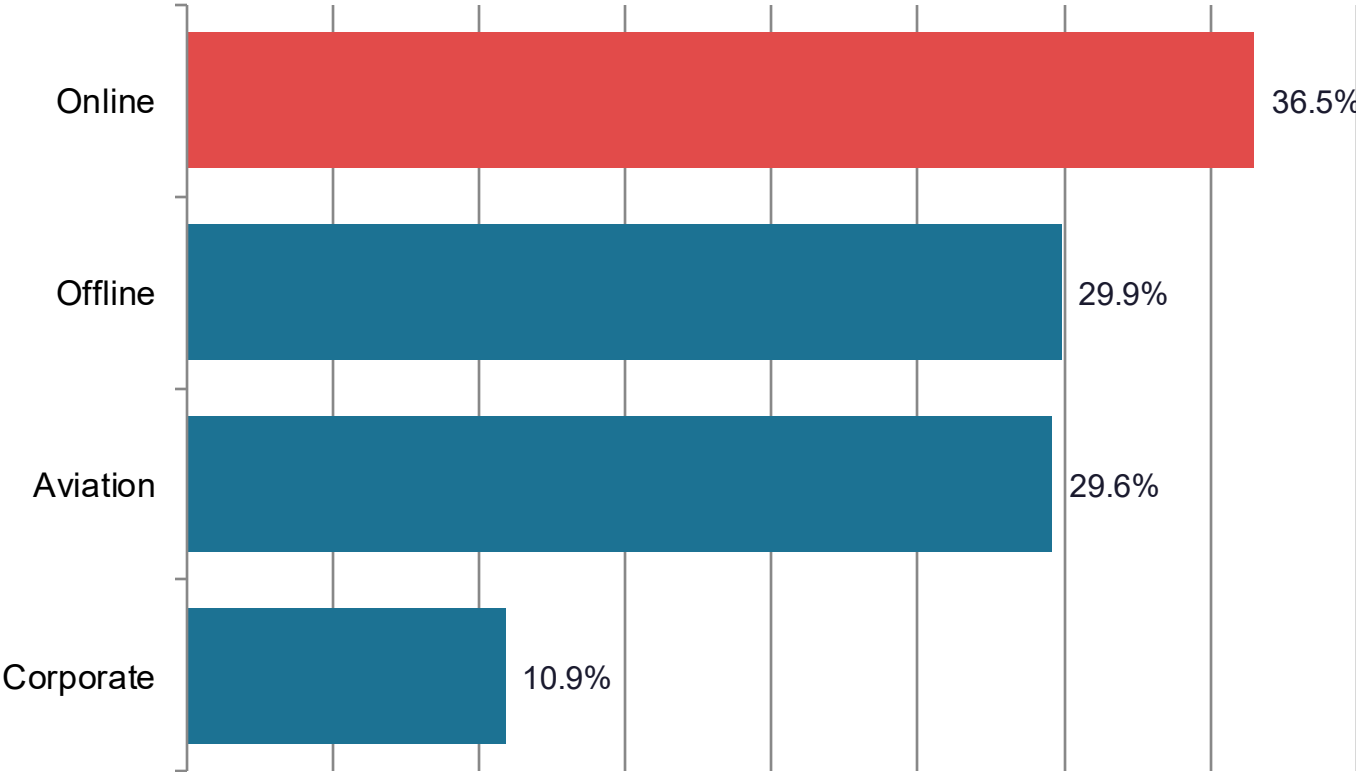
Reactive Staff

Teams find out a booking is gone only when the guest is already gone.

WHAT THE DATA SHOWS

Not all bookings carry the same risk.

Cancellation rate by booking channel



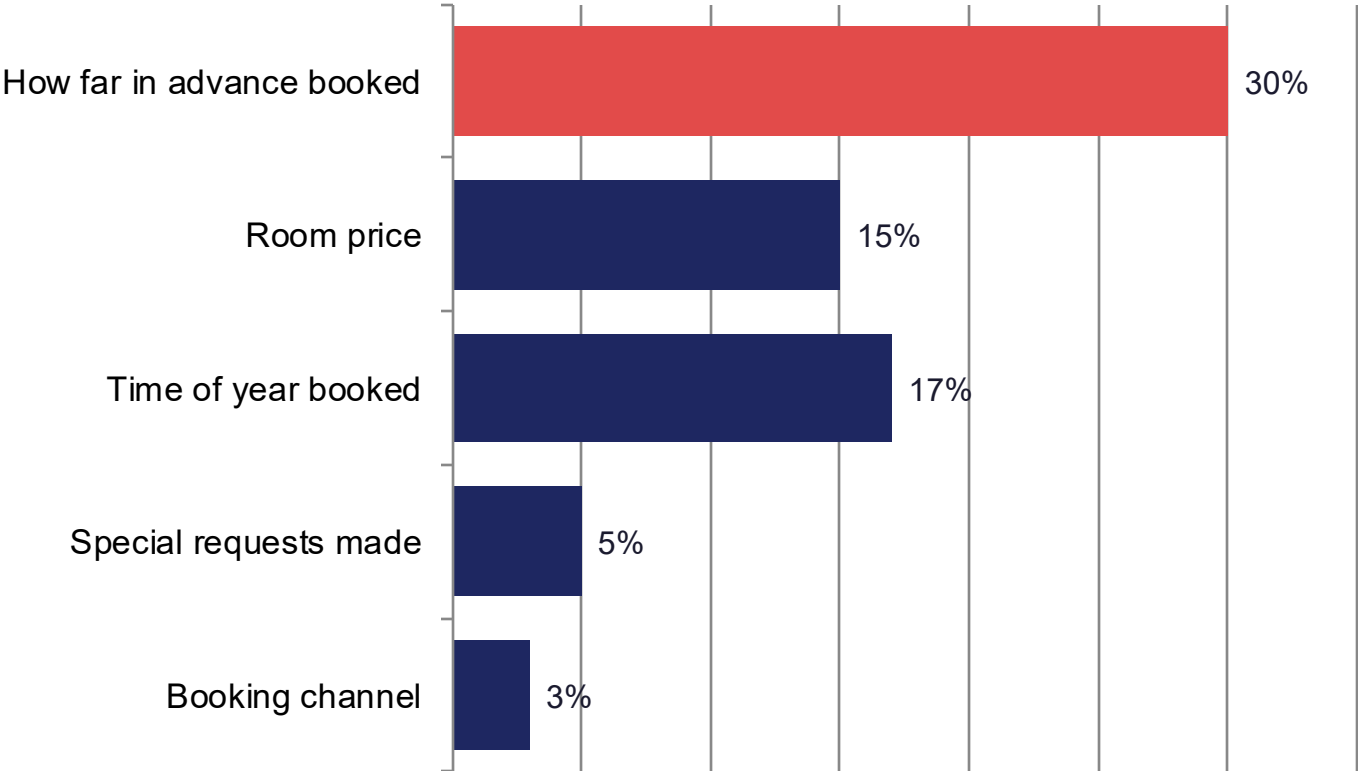
What this tells the business

Online bookings cancel over 3× more often than Corporate bookings. The channel a guest books through is one of the clearest, most actionable risk signals available — and it costs nothing extra to know.

WHAT THE DATA SHOWS

Five factors explain most of the risk.

Share of the system's decision-making, by factor

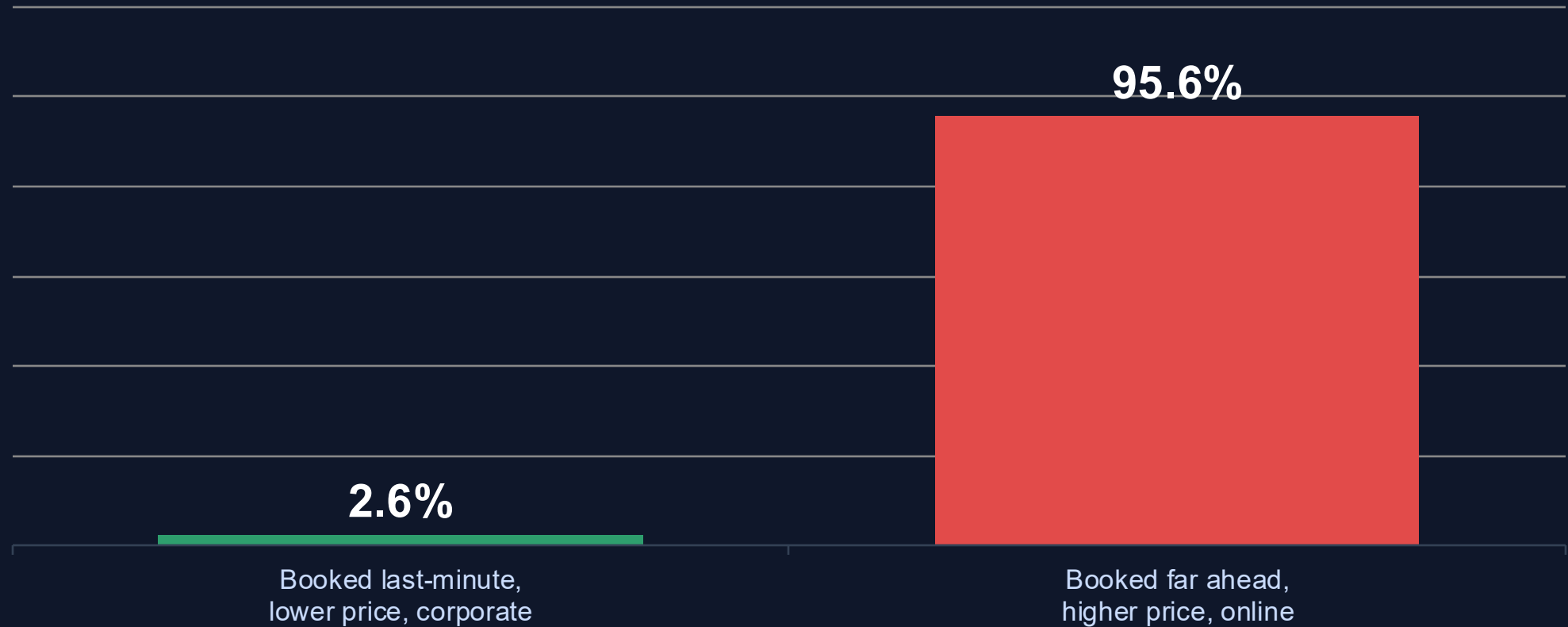


What this tells the business

How far in advance a booking was made is, by far, the single biggest driver of cancellation risk — twice as influential as room price. Long-lead bookings deserve the closest attention.

SEEING IT IN ACTION

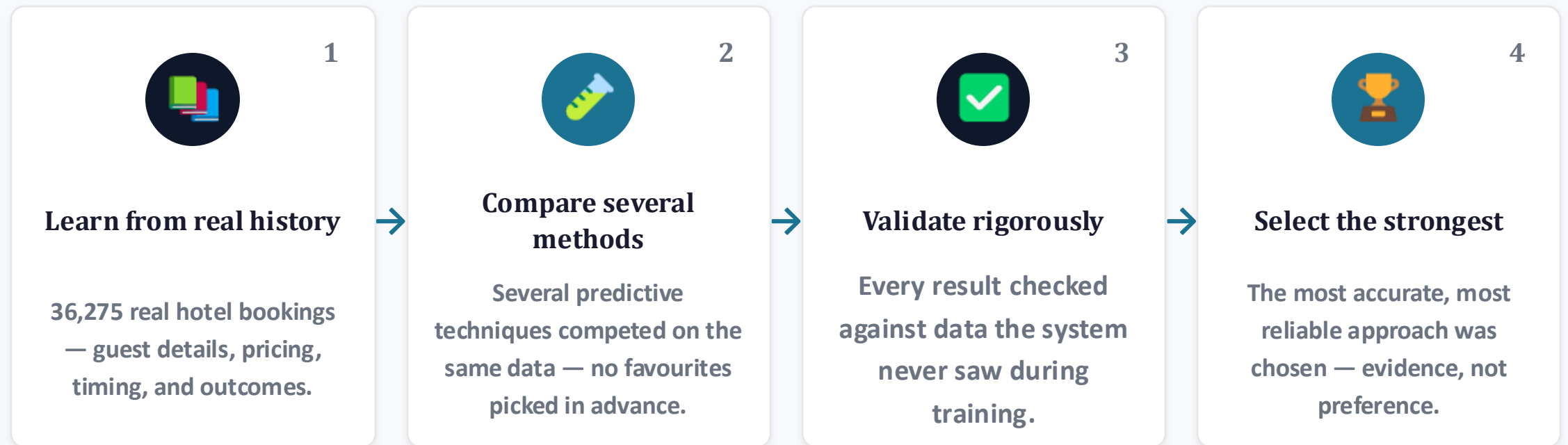
Two real bookings. Two very different risk levels.



Same predictive system, two real booking profiles — the risk score moves from near-certain to keep, to near-certain to cancel.

OUR APPROACH

Built the way good forecasting should be: tested, not assumed.



RESULTS THAT MATTER

Tested against thousands of real bookings the system never saw.

90%

Overall Forecasting
Accuracy

82%

of Actual Cancellations
Caught in Advance

96%

Reliability at Ranking
High- vs. Low-Risk Bookings

Several methods were compared head-to-head, and the strongest performer was chosen on evidence — not assumption.

THE EXPERIENCE

From prediction to action, in one screen.

<https://hotel-cancellation-prediction-7cn3s7u84sjrjkhcgki4r4.streamlit.app/>



Live Risk Score

A clear, color-coded read on every booking's cancellation risk — Low, Medium, or High.



Plain-Language Reasoning

The 'why' behind every score, explained the way a human analyst would explain it.



Recommended Actions

Ranked next steps — from a reminder email to a deposit request — matched to the risk level.



Financial Impact View

Revenue at risk, and the payoff of intervening, laid out before a single action is taken.

RECOMMENDATIONS

What the hotel should actually do with this.

- 1 Watch long-lead bookings closely**
Send a confirmation reminder or request a deposit for reservations made far in advance — the single biggest risk factor.
- 2 Add friction to online bookings**
Card verification or a rate-locked non-refundable option for online channels, where cancellations run highest.
- 3 Reward commitment signals**
Guests who make special requests or have stayed before cancel far less — prioritize them lower for follow-up.
- 4 Use the risk score to guide effort**
Focus staff time and incentives on the bookings the system flags as high-risk, not every booking equally.

BUSINESS IMPACT

**A small intervention, applied at the right moment,
protects a lot of revenue.**

WITHOUT EARLY WARNING



**A high-value booking cancels without notice.
The room sits empty. Revenue is gone,
and there was no chance to react.**

WITH StayCertain



**The same booking is flagged as high-risk weeks
earlier. A simple reminder or deposit request
is sent — the stay is protected before it's lost.**

Every cancellation predicted is revenue protected.

Predict the future. Prevent the loss.
StayCertain

Thank you — questions welcome.